

# Background: linear, quadratic and nonlinear programming

# Outline

## Linear programming theory

### Duality

The optimality conditions: complementary slackness

Sensitivity analysis and shadow prices

## Nonlinear optimization theory

Convexity and quadratic programming

The optimality conditions: the KKT conditions

## The sensitivity analysis protocol

# What this chapter is about

- ▶ We collect the **theoretical tools** used throughout the course
- ▶ The *theory of linear programming*: duality, optimality conditions, sensitivity analysis
- ▶ Its *nonlinear extension*: convexity, KKT conditions
- ▶ It closes with the **sensitivity protocol** used in every laboratory session
- ▶ It also serves as a quick reference

# Linear programming theory

# Introduction: Linear programming theory

- ▶ Every maximization LP has a **dual** minimization problem
- ▶ The dual variables are the **shadow prices**: the value of one extra unit of resource
- ▶ **Complementary slackness** tells us when a primal-dual pair is optimal
- ▶ **Sensitivity analysis** tells us how the optimum changes when the data change: the question a decision maker cares about

# Sets and data

- ▶  $M = \{1, 2, \dots, m\}$ : constraint indices;  $N = \{1, 2, \dots, n\}$ : variable indices
- ▶ **Data**: costs  $c_j \in \mathbb{R}$  ( $j \in N$ ), coefficients  $a_{ij} \in \mathbb{R}$ , right-hand sides  $b_i \in \mathbb{R}$  ( $i \in M$ )
- ▶  $M_{\leq}$ ,  $M_{=}$ ,  $M_{\geq}$  partition  $M$  according to the direction of the constraint
- ▶  $N_{\geq 0}$ ,  $N_{\geq 0}$ ,  $N_{\leq 0}$  partition  $N$  according to the sign of the variable (non-negative, free, non-positive)
- ▶ **Variables**: the  $x_j$  of the primal and, in the dual, one  $\pi_i$  for each of the  $m$  constraints of the primal

# The primal (P)

## The primal (P), general form

$$\begin{aligned} \text{(P)} \quad & \max \sum_{j \in N} c_j x_j \\ \text{subject to} \quad & \sum_{j \in N} a_{ij} x_j \leq b_i, \quad \forall i \in M_{\leq}, \\ & \sum_{j \in N} a_{ij} x_j = b_i, \quad \forall i \in M_{=}, \\ & \sum_{j \in N} a_{ij} x_j \geq b_i, \quad \forall i \in M_{\geq}, \\ & x_j \geq 0, \quad \forall j \in N_{\geq 0}, \\ & x_j \leq 0, \quad \forall j \in N_{\leq 0}, \\ & x_j \text{ free}, \quad \forall j \in N_{\text{free}}. \end{aligned}$$

## The primal (P) (2/2)

- ▶ Three constraint directions ( $\leq$ ,  $=$ ,  $\geq$ ) and three variable signs (non-negative, free, non-positive)

# The dual (D)

## The dual (D), general form

$$\begin{aligned} \text{(D)} \quad & \min \sum_{i \in M} b_i \pi_i \\ & \text{subject to} \quad \sum_{i \in M} a_{ij} \pi_i \geq c_j, \quad \forall j \in N_{\geq 0}, \\ & \quad \quad \quad \sum_{i \in M} a_{ij} \pi_i = c_j, \quad \forall j \in N_{=0}, \\ & \quad \quad \quad \sum_{i \in M} a_{ij} \pi_i \leq c_j, \quad \forall j \in N_{\leq 0}, \\ & \quad \quad \quad \pi_i \geq 0, \quad \forall i \in M_{\leq}, \\ & \quad \quad \quad \pi_i \geq 0, \quad \forall i \in M_{=}, \\ & \quad \quad \quad \pi_i \leq 0, \quad \forall i \in M_{\geq}. \end{aligned}$$

## The dual (D) (2/2)

- ▶ The sign of  $\pi_i$  depends on the direction of constraint  $i$  of the primal

# The rules of duality (1/2)

Every constraint of the primal generates a dual variable, every variable of the primal a dual constraint:

| Primal (max)                                 | Dual (min)              |
|----------------------------------------------|-------------------------|
| constraint $\leq b_i$ ( $i \in M_{\leq}$ )   | variable $\pi_i \geq 0$ |
| constraint $= b_i$ ( $i \in M_{=}$ )         | variable $\pi_i \geq 0$ |
| constraint $\geq b_i$ ( $i \in M_{\geq}$ )   | variable $\pi_i \leq 0$ |
| variable $x_j \geq 0$ ( $j \in N_{\geq 0}$ ) | constraint $\geq c_j$   |
| variable $x_j \geq 0$ ( $j \in N_{\geq 0}$ ) | constraint $= c_j$      |
| variable $x_j \leq 0$ ( $j \in N_{\leq 0}$ ) | constraint $\leq c_j$   |

# The rules of duality (2/2)

Two quick readings:

- ▶ the *equalities* of the primal give *free* dual variables (and vice versa)
- ▶ in a maximization a resource constraint  $\leq$  has  $\pi_i \geq 0$  (more resource, more value)

For a *minimization* primal the table is read from right to left

# The duality theorems

- ▶ The primal-dual pair is linked by the **two duality theorems**
- ▶ In the first, the *bar* denotes any feasible solution
- ▶ In the second, the *tilde* denotes an optimal solution

## Weak duality theorem

For every feasible solution  $(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n)$  of the primal (P) and every feasible solution  $(\bar{\pi}_1, \bar{\pi}_2, \dots, \bar{\pi}_m)$  of the dual (D) we have

$$\sum_{j \in N} c_j \bar{x}_j \leq \sum_{i \in M} b_i \bar{\pi}_i.$$

# Strong duality

## Strong duality theorem

The primal (P) has an optimal solution  $(\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n)$  *if and only if* the dual (D) has an optimal solution  $(\tilde{\pi}_1, \tilde{\pi}_2, \dots, \tilde{\pi}_m)$ , and in that case the two optimal values coincide:

$$\sum_{j \in N} c_j \tilde{x}_j = \sum_{i \in M} b_i \tilde{\pi}_i.$$

# What the two theorems say

- ▶ **Weak duality:** every feasible solution of the dual is an *upper bound* for the value of the maximization primal (and every feasible solution of the primal a lower bound for the dual)
- ▶ If two feasible solutions have the same value, they are both optimal
- ▶ **Strong duality:** at the optimum this always happens, the gap closes
- ▶ Existence does not mean uniqueness: there may be more than one optimal solution, but the optimal values coincide

## Example: the dual of the $2 \times 2$ LP (1/3)

Let us apply the rules to the  $2 \times 2$  LP, which is exactly in the canonical maximization form (constraints  $\leq$ , non-negative variables):

$$\begin{array}{ll}\max & 30 x_1 + 50 x_2 \\ \text{subject to} & x_1 + 3 x_2 \leq 90, \\ & 2 x_1 + x_2 \leq 80, \\ & x_1, x_2 \geq 0.\end{array}$$

## Example: the dual of the $2 \times 2$ LP (2/3)

- ▶ **Step 1 — every constraint generates a dual variable.**  
Two constraints  $\leq$  in a maximization  $\Rightarrow \pi_1, \pi_2 \geq 0$
- ▶ **Step 2 — every variable generates a dual constraint.**  
The coefficients are the *columns* of the primal, the right-hand sides are the costs:
  - ▶ from  $x_1 \geq 0$  we get  $\pi_1 + 2\pi_2 \geq 30$
  - ▶ from  $x_2 \geq 0$  we get  $3\pi_1 + \pi_2 \geq 50$
- ▶ **Step 3 — the objective swaps the roles:** the right-hand sides (90, 80) become the costs of the dual, which is a *minimization* problem

## Example: the dual of the $2 \times 2$ LP (3/3)

The dual is:

$$\begin{array}{ll} \min & 90 \pi_1 + 80 \pi_2 \\ \text{subject to} & \pi_1 + 2 \pi_2 \geq 30, \\ & 3 \pi_1 + \pi_2 \geq 50, \\ & \pi_1, \pi_2 \geq 0. \end{array}$$

## Example: all the cases in a single LP (1/3)

A *minimization* LP with all the cases of the table: three variables with the three signs and three constraints with the three directions (the table is read from right to left):

$$\begin{array}{llll} \min & 5x_1 + 8x_2 - 9x_3 & & \\ \text{subject to} & x_1 + x_2 & \geq & 30, \\ & x_1 + x_2 - x_3 & = & 100, \\ & x_1 - 2x_2 & \leq & -20, \\ & x_1 & \geq & 0, \\ & & x_2 & \begin{array}{l} \geq \\ \leq \end{array} & 0, \\ & & & x_3 \leq & 0. \end{array}$$

## Example: all the cases in a single LP (2/3)

- ▶ **Step 1 — the dual variables**, one per constraint, with the sign given by the direction:
  - ▶  $\pi_1 \geq 0$  (constraint  $\geq$  in a minimization)
  - ▶  $\pi_2 \stackrel{=}{=} 0$  (equality)
  - ▶  $\pi_3 \leq 0$  (constraint  $\leq$  in a minimization)
- ▶ **Step 2 — the dual constraints**, one per variable, with the direction given by the sign:
  - ▶ from  $x_1 \geq 0$  we get  $\pi_1 + \pi_2 + \pi_3 \leq 5$
  - ▶ from  $x_2 \stackrel{=}{=} 0$  we get  $\pi_1 + \pi_2 - 2\pi_3 = 8$
  - ▶ from  $x_3 \leq 0$  we get  $-\pi_2 \geq -9$
- ▶ **Step 3 — the objective**: the dual is a maximization,  
 $\max 30\pi_1 + 100\pi_2 - 20\pi_3$

## Example: all the cases in a single LP (3/3)

The dual is:

$$\begin{array}{llllll} \max & 30 \pi_1 & + & 100 \pi_2 & - & 20 \pi_3 \\ \text{subject to} & \pi_1 & + & \pi_2 & + & \pi_3 & \leq & 5, \\ & \pi_1 & + & \pi_2 & - & 2 \pi_3 & = & 8, \\ & & & - & \pi_2 & & \geq & -9, \\ & \pi_1 & & & & & \geq & 0, \\ & & & & \pi_2 & & \geq & 0, \\ & & & & & & \leq & 0, \\ & & & & & \pi_3 & \leq & 0. \end{array}$$

# Slacks and reduced costs

To every primal constraint we associate its *slack*  $\bar{s}_i$ , and to every variable its *reduced cost*  $\bar{c}_j$ , that is the slack of the corresponding *dual* constraint:

$$\bar{s}_i = \sum_{j \in N} a_{ij} \bar{x}_j - b_i, \quad \forall i \in M, \quad \bar{c}_j = c_j - \sum_{i \in M} a_{ij} \bar{\pi}_i, \quad \forall j \in N$$

- ▶ The slack measures how far constraint  $i$  is from being binding
- ▶ The reduced cost measures the coefficient  $c_j$  net of the value of the resources consumed by  $x_j$ , evaluated at the dual variables

# Optimality conditions

## LP optimality conditions (complementary slackness)

A pair of feasible solutions,  $(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n)$  for the primal (P) and  $(\bar{\pi}_1, \bar{\pi}_2, \dots, \bar{\pi}_m)$  for the dual (D), is optimal for both problems *if and only if*

$$\bar{\pi}_i \cdot \bar{s}_i = 0, \quad \forall i \in M,$$

$$\bar{x}_j \cdot \bar{c}_j = 0, \quad \forall j \in N.$$

# Complementary slackness in words

- ▶ **Slack constraint  $\Rightarrow$  dual at zero**

$$\bar{s}_i \neq 0 \Rightarrow \bar{\pi}_i = 0$$

- ▶ **Variable different from zero  $\Rightarrow$  null reduced cost**

$$\bar{x}_j \neq 0 \Rightarrow \bar{c}_j = 0$$

- ▶ At the optimum these are *implications*, not equivalences: because of degeneracy an active constraint may have a null dual, and a variable at zero a null reduced cost
- ▶ The verification examples further on check exactly these conditions

## Two readings for the laboratory (1/3)

**Shadow price**  $\tilde{\pi}_i$ : the marginal value of the right-hand side  $b_i$ ,

$$\tilde{\pi}_i = \frac{\partial \tilde{z}}{\partial b_i}$$

- ▶ It holds for  $b_i$  within the *validity range of the right-hand side*  $[b_i^{\min}, b_i^{\max}]$
- ▶ It is the largest interval in which, varying only  $b_i$ , the shadow price remains exactly  $\tilde{\pi}_i$  (beyond it, it changes and must be recomputed)
- ▶ Special case: inactive constraint  $\Rightarrow \tilde{\pi}_i = 0$

## Two readings for the laboratory (2/3)

**Reduced cost  $\bar{c}_j$**  (the slack of the dual constraint):

- ▶ for a variable at zero it says by how much  $c_j$  must improve for it to become worth activating
- ▶ it holds for  $c_j$  within the *validity range of the coefficient*  $[c_j^{\min}, c_j^{\max}]$ : the largest interval in which, varying only  $c_j$ , the optimal solution does not change
- ▶ for a variable at zero the profitability threshold is precisely one of the two endpoints
- ▶ special case: variable different from zero  $\Rightarrow \bar{c}_j = 0$

## Two readings for the laboratory (3/3)

### Beware of degeneracy:

- ▶ there may exist a variable *at zero with null reduced cost*, even though it is not used
- ▶ hence only the implication holds,

$$\bar{c}_j \neq 0 \Rightarrow \text{variable on one of its bounds}$$

not the converse

## The signs of shadow prices (1/4)

The shadow price is always, with a single convention,

$$\tilde{\pi}_i = \frac{\partial \tilde{z}}{\partial b_i}$$

the derivative of the optimal value with respect to the right-hand side of constraint  $i$ : by how much  $\tilde{z}$  changes if  $b_i$  increases by one (small) unit

The sign is always deduced by answering **two questions**

## The signs of shadow prices (2/4)

- ▶ **Does increasing  $b_i$  enlarge or shrink the feasible region?**
  - ▶ with a constraint  $\leq$  it enlarges it (the ceiling is raised)
  - ▶ with a constraint  $\geq$  it shrinks it (the requirement grows)
  - ▶ with an equality it merely shifts it
- ▶ **How does a wider region change the optimum?**
  - ▶ it can never worsen it
  - ▶ in a minimization the optimal value decreases or stays the same
  - ▶ in a maximization it increases or stays the same

## The signs of shadow prices (3/4)

Combining the two answers:

| Direction of the constraint                         | minimization           | maximization           |
|-----------------------------------------------------|------------------------|------------------------|
| $\leq$ ( $b_i$ grows $\Rightarrow$ wider region)    | $\tilde{\pi}_i \leq 0$ | $\tilde{\pi}_i \geq 0$ |
| $\geq$ ( $b_i$ grows $\Rightarrow$ narrower region) | $\tilde{\pi}_i \geq 0$ | $\tilde{\pi}_i \leq 0$ |
| $=$                                                 | any sign (free dual)   |                        |

# The signs of shadow prices (4/4)

Two typical readings in a *minimization*:

- ▶ a constraint  $\leq$  (a capacity) has  $\tilde{\pi}_i \leq 0$ : relaxing it reduces the cost, and  $|\tilde{\pi}_i|$  is the saving for each extra unit of capacity
- ▶ a constraint  $\geq$  (a requirement to be met) has  $\tilde{\pi}_i \geq 0$ : raising the requirement costs, and  $\tilde{\pi}_i$  is the marginal cost

In a *maximization* the readings are reversed

In every case an inactive constraint has  $\tilde{\pi}_i = 0$  by complementarity

## Example: duality on the $2 \times 2$ LP (1/4)

Let us take up again the  $2 \times 2$  LP and its dual and verify the optimality conditions.

**Step 1 — solving we obtain:**

$$(\tilde{x}_1, \tilde{x}_2) = (30, 20), \quad \tilde{z} = 1900,$$

$$(\tilde{\pi}_1, \tilde{\pi}_2) = (14, 8), \quad (\bar{c}_1, \bar{c}_2) = (0, 0)$$

with validity ranges

$$[b_1^{\min}, b_1^{\max}] = [40, 240], \quad [b_2^{\min}, b_2^{\max}] = [30, 180],$$

$$[c_1^{\min}, c_1^{\max}] = [50/3, 100], \quad [c_2^{\min}, c_2^{\max}] = [15, 90].$$

Both constraints are active.

## Example: duality on the $2 \times 2$ LP (2/4)

### Step 2 — verification of the shadow prices.

- ▶ One extra unit of  $b_1$  is worth 14, one extra unit of  $b_2$  is worth 8
- ▶  $\tilde{x}_1$  and  $\tilde{x}_2$  are different from zero: by complementary slackness their dual constraints must be active ( $\bar{c}_1 = \bar{c}_2 = 0$ ):

$$\tilde{\pi}_1 + 2\tilde{\pi}_2 = 14 + 16 = 30 = c_1, \quad 3\tilde{\pi}_1 + \tilde{\pi}_2 = 42 + 8 = 50 = c_2 \quad \checkmark$$

## Example: duality on the $2 \times 2$ LP (3/4)

**Step 3 — verification of strong duality.**

$$\sum_{i \in M} b_i \tilde{\pi}_i = 90 \cdot 14 + 80 \cdot 8 = 1260 + 640 = 1900 = \tilde{z} \quad \checkmark$$

## Example: duality on the $2 \times 2$ LP (4/4)

**Step 4 — verification of the reduced costs.** With

$$\bar{c}_j = c_j - \sum_{i \in M} a_{ij} \tilde{\pi}_i:$$

$$\bar{c}_1 = 30 - (1 \cdot 14 + 2 \cdot 8) = 0, \quad \bar{c}_2 = 50 - (3 \cdot 14 + 1 \cdot 8) = 0$$

- ▶ Both null, as for every positive variable at the optimum: the two variables repay exactly the resources they consume
- ▶ Only a variable sitting on one of its bounds can have a reduced cost different from zero

## Example: verification on all the cases (1/5)

Let us take up again the “all the cases” LP and its dual: the cases are the same ones that will come back, one at a time, in the application chapters.

**Step 1 — solving we obtain:**

$$(\tilde{x}_1, \tilde{x}_2, \tilde{x}_3) = (60, 40, 0), \quad \tilde{z} = 620,$$

$$(\tilde{\pi}_1, \tilde{\pi}_2, \tilde{\pi}_3) = (0, 6, -1), \quad (\bar{c}_1, \bar{c}_2, \bar{c}_3) = (0, 0, -3)$$

- ▶ ranges of the right-hand sides:  $(-\infty, 100]$ ,  $[30, +\infty)$ ,  $[-200, +\infty)$
- ▶ ranges of the coefficients:  $(-\infty, 8]$ ,  $[5, 17]$ ,  $(-\infty, -6]$
- ▶ the second and third constraints are active, with  $\tilde{x}_3 = 0$

## Example: verification on all the cases (2/5)

### Step 2 — shadow prices with the rules of the table.

- ▶  $\tilde{\pi}_1 = 0$ , consistent with complementarity: the first constraint is slack ( $\tilde{x}_1 + \tilde{x}_2 = 100 > 30$ )
- ▶  $\tilde{\pi}_2 = 6$  (free dual variable, here positive)
- ▶  $\tilde{\pi}_3 = -1 \leq 0$ , as it must be for a constraint  $\leq$  in a minimization

## Example: verification on all the cases (3/5)

The dual constraints of  $x_1$  and  $x_2$  (different from zero) must be active:

$$\underbrace{0 + 6 - 1}_{x_1} = 5 = c_1, \quad \underbrace{0 + 6 - 2 \cdot (-1)}_{x_2} = 8 = c_2 \quad \checkmark$$

( $x_2$  is free: its dual constraint is an equality)

Verification by perturbation:

$$b_2 = 101 \Rightarrow \tilde{z} = 626 (+6), \quad b_3 = -19 \Rightarrow \tilde{z} = 619 (-1) \quad \checkmark$$

## Example: verification on all the cases (4/5)

**Step 3 — verification of strong duality.**

$$\sum_{i \in M} b_i \tilde{\pi}_i = 30 \cdot 0 + 100 \cdot 6 + (-20) \cdot (-1) = 620 = \tilde{z} \quad \checkmark$$

**Step 4 — reduced costs, one per sign.** With

$$\bar{c}_j = c_j - \sum_{i \in M} a_{ij} \tilde{\pi}_i:$$

$$\bar{c}_1 = 5 - (0 + 6 - 1) = 0, \quad \bar{c}_2 = 8 - (0 + 6 + 2) = 0$$

consistent with  $\tilde{x}_1, \tilde{x}_2 \neq 0$

## Example: verification on all the cases (5/5)

$x_3$  equals zero, which for a variable  $\leq 0$  is its *upper bound*:

$$\bar{c}_3 = -9 - (-1) \cdot 6 = -3$$

- ▶ forcing  $x_3 = -1$  would worsen the optimum by 3
- ▶  $x_3$  would become active only if its coefficient rose to at least  $-6$  (precisely the endpoint  $c_3^{\max}$  of the validity range)

Verification:

$$x_3 = -1 \Rightarrow \tilde{z} = 623 (+3) \quad \checkmark$$

# Sensitivity analysis

- ▶ Solving an LP gives an optimal solution
- ▶ **Sensitivity analysis** answers the next question, the one that really matters to a decision maker:  
*how does the optimum change if the data change?*
- ▶ In LPs a large part of the answer comes for free together with the solution, in the form of **four numbers** for each constraint and variable

# The four numbers (1/2)

- ▶ The **shadow price**  $\tilde{\pi}_i$  of a constraint: by how much the optimal value improves if the right-hand side of the constraint increases by one unit  
“how much I would pay for one extra unit of that resource”
- ▶ The **validity range of the right-hand side**  $[b_i^{\min}, b_i^{\max}]$ : where that shadow price remains exact (outside it, the optimization must be redone)

## The four numbers (2/2)

- ▶ The **reduced cost**  $\bar{c}_j$  of a variable at zero: by how much its objective coefficient must improve for it to become worth activating
- ▶ The **validity range of the objective coefficient**  $[c_j^{\min}, c_j^{\max}]$ : where the coefficient can vary without the optimal solution changing
- ▶ In the  $2 \times 2$  LP:  $c_1$  can vary in  $[50/3 \approx 16.7, 100]$  and  $c_2$  in  $[15, 90]$  without moving the solution  $(30, 20)$

## Example: shadow prices in practice (1/2)

$2 \times 2$  LP: optimum 1900 with duals  $\tilde{\pi}_1 = 14$  and  $\tilde{\pi}_2 = 8$ .

**Reading.** One extra unit of  $b_1$  makes the optimum grow by 14; if obtaining it cost 12, it would be worth it ( $14 > 12$ ).

**Verification by perturbation.** Re-solving:

$$b_1 = 91 \Rightarrow 1914 = 1900 + 14, \quad b_1 = 100 \Rightarrow 2040 = 1900 + 10 \cdot 14$$

the shadow price is still exact because 100 lies inside the validity range  $[40, 240]$ .

## Example: shadow prices in practice (2/2)

### Where it stops holding.

- ▶ At  $b_1 = 240$  the first constraint stops being the scarce resource
- ▶ Beyond that, every additional unit is worth less than 14 and everything must be recomputed
- ▶ This is why every shadow price is quoted *together* with its validity range

### Inactive constraints.

- ▶ If we had  $b_1 = 300$  (constraint inactive at the optimum), its shadow price would be 0
- ▶ A resource left over is worth nothing at the margin

## Example: is a third variable worth it? (1/3)

- ▶ In the  $2 \times 2$  LP both variables are positive at the optimum and their reduced costs equal 0: this is always the case for variables different from zero
- ▶ To see a reduced cost at work we need a variable at zero
- ▶ Let us add to the model a variable  $x_3 \geq 0$  with coefficient  $c_3 = 20$  in the objective and consumptions  $a_{13} = a_{23} = 1$  in the two constraints

## Example: is a third variable worth it? (2/3)

**Step 1 — valuation with the shadow prices.** The resources absorbed by one unit of  $x_3$  are worth, at the shadow prices,

$$1 \cdot 14 + 1 \cdot 8 = 22 > 20 = c_3$$

every unit of  $x_3$  would take resources away from better uses.

**Step 2 — the reduced cost.** The difference is precisely the reduced cost:

$$\bar{c}_3 = 20 - 22 = -2$$

Solving confirms it: the optimal solution remains  $(30, 20, 0)$  with  $\tilde{z} = 1900$ ,  $x_3$  is at zero with  $\bar{c}_3 = -2$  and  $c_3^{\max} = 22$ .

## Example: is a third variable worth it? (3/3)

- ▶ The coefficient  $c_3$  would have to rise to at least 22 for it to be worth activating  $x_3$
- ▶ **Step 3 — counter-check.** With  $c_3 = 23 > 22$  the solution really does change, and not by a little: the new optimal solution is  $(0, 5, 75)$  with value 1975
- ▶ The reduced cost tells us *when* a variable at zero enters the solution, not *how* the solution changes: that has to be recomputed

# Summary: Linear programming theory (1/2)

## Duality:

- ▶ Every constraint generates a dual variable, every variable a dual constraint
- ▶ Weak duality: the dual gives a bound; strong duality: at the optimum the values coincide

## Complementary slackness:

- ▶  $\bar{\pi}_i \cdot \bar{s}_i = 0$  and  $\bar{x}_j \cdot \bar{c}_j = 0$
- ▶ Slack constraint  $\Rightarrow$  null dual; non-null variable  $\Rightarrow$  null reduced cost

# Summary: Linear programming theory (2/2)

## Sensitivity:

- ▶ Shadow price  $\tilde{\pi}_i = \partial \tilde{z} / \partial b_i$ , valid within its range  $[b_i^{\min}, b_i^{\max}]$
- ▶ The sign is deduced from: does the region widen or narrow?  
Min or max?
- ▶ Reduced cost: by how much  $c_j$  must improve for a variable at zero to be worth it
- ▶ A number is always quoted *together* with its validity range

# Nonlinear optimization theory



# Introduction: Nonlinear optimization

- ▶ We extend the tools of LP to **quadratic** and **nonlinear** problems
- ▶ **Convexity** is the property that allows us to certify the optimum
- ▶ The **KKT conditions** generalize complementary slackness
- ▶ The KKT multipliers generalize the shadow prices: we will still have an economic reading

# Convex problems

- ▶ A problem  $\min f(x_1, x_2, \dots, x_n)$  over a feasible set  $X$  is **convex** if
  - ▶  $f$  is convex
  - ▶  $X$  is convex
- ▶ Let us see the property we shall use all the time

# Why convexity matters

- ▶ In a convex problem every local minimum is global: optimality can be *certified*
- ▶ In non-convex problems (for instance a pricing problem with a bilinear term  $p \cdot q$ ) a local method guarantees only a local optimum
- ▶ Non-convex QPs can nonetheless be solved to global optimality with dedicated techniques, but the computational cost grows

# Quadratic programming

- ▶ A **QP** has objective

$$\frac{1}{2} \sum_i \sum_j q_{ij} x_i x_j + \sum_j c_j x_j$$

and linear constraints

- ▶ It is convex if and only if the matrix  $\mathbf{Q} = (q_{ij})$  is positive semidefinite ( $\mathbf{Q} \succeq 0$ )
- ▶ In this course QPs appear every time a cost or a risk grows more than linearly with the decision

## Example: a $2 \times 2$ QP (1/3)

A minimization QP with a separable quadratic objective (the second variable “costs” twice as much as the first, at the margin) and a threshold constraint on the sum:

$$\begin{aligned} \min \quad & x_1^2 + 2x_2^2 \\ \text{subject to} \quad & x_1 + x_2 \geq 6, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Here  $\mathbf{Q} = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix} \succ 0$ : the QP is convex and the optimum found is global.

## Example: a $2 \times 2$ QP (2/3)

**Step 1 — solving we obtain:**

$$\tilde{x}_1 = 4, \quad \tilde{x}_2 = 2, \quad \tilde{f} = 24, \quad \lambda = 8$$

**Step 2 — verification of the shadow price.**

- ▶ The constraint is active ( $4 + 2 = 6$ ): the objective grows with the variables, hence the sum does not exceed what is needed
- ▶ The partial derivatives coincide:

$$2\tilde{x}_1 = 8 = 4\tilde{x}_2 = \lambda \quad \checkmark$$

- ▶ If they were different it would be worth shifting quantity from one variable to the other; the common derivative is precisely the multiplier

## Example: a $2 \times 2$ QP (3/3)

### Step 3 — verification by perturbation.

With right-hand side 7 we obtain  $(\tilde{x}_1, \tilde{x}_2) = (14/3, 7/3)$  with

$$\tilde{f} = 32.67 \approx 24 + 8$$

- ▶ On top of that there is the curvature term
- ▶ Here the shadow price is exact only at the margin, not for finite increments

# The Lagrangian

- ▶ The optimality conditions of constrained nonlinear problems are known as the **KKT conditions** (after Karush, Kuhn and Tucker)
- ▶ Problem:  $\min f(x_1, \dots, x_n)$  subject to
  - ▶  $g_i(x_1, \dots, x_n) \leq 0$  for every  $i \in \{1, 2, \dots, m\}$
  - ▶  $h_j(x_1, \dots, x_n) = 0$  for every  $j \in \{1, 2, \dots, q\}$
- ▶ The **Lagrangian** is

$$L = f(x_1, \dots, x_n) + \sum_{i=1}^m \lambda_i g_i(x_1, \dots, x_n) + \sum_{j=1}^q \nu_j h_j(x_1, \dots, x_n), \quad \lambda_i \geq 0$$

# The KKT conditions

## KKT conditions

At a regular optimal point  $(\tilde{x}_1, \dots, \tilde{x}_n)$  there exist multipliers  $\lambda_i \geq 0$  and  $\nu_j$  such that

$$\frac{\partial L}{\partial x_k}(\tilde{x}_1, \dots, \tilde{x}_n) = 0 \quad \forall k \in \{1, 2, \dots, n\} \quad (\text{stationarity}),$$

$$\lambda_i g_i(\tilde{x}_1, \dots, \tilde{x}_n) = 0 \quad \forall i \in \{1, 2, \dots, m\} \quad (\text{complementarity}).$$

If the problem is convex, the KKT conditions are also *sufficient*: a point that satisfies them is a global optimum.

# The KKT conditions are complementary slackness, generalized

- ▶ The two parts of the chapter mirror each other: for an LP the KKT conditions reduce *exactly* to complementary slackness
- ▶ The stationarity of the Lagrangian is equivalent to the feasibility of the dual:
  - ▶ the multipliers  $\lambda_i$  are the dual variables  $\pi_i$
  - ▶ the multipliers of the sign constraints are the reduced costs
- ▶ The complementarity  $\lambda_i g_i(\tilde{x}_1, \dots, \tilde{x}_n) = 0$  is the complementary slackness condition  $\bar{\pi}_i \cdot \bar{s}_i = 0$
- ▶ Complementary slackness is the KKT conditions in the linear case

## Example: the KKT conditions on the QP (1/3)

Let us take up again the QP just solved:  $\min x_1^2 + 2x_2^2$  subject to  $x_1 + x_2 \geq 6$ ,  $x_1, x_2 \geq 0$ .

**Step 1.** The constraint in standard form:

$$g(x_1, x_2) = 6 - x_1 - x_2 \leq 0$$

**Step 2.** Stationarity of the Lagrangian

$$L = x_1^2 + 2x_2^2 + \lambda(6 - x_1 - x_2):$$

$$\frac{\partial L}{\partial x_1} = 2x_1 - \lambda = 0, \quad \frac{\partial L}{\partial x_2} = 4x_2 - \lambda = 0 \Rightarrow x_1 = \frac{\lambda}{2}, \quad x_2 = \frac{\lambda}{4}$$

(the constraints  $x_1, x_2 \geq 0$  will turn out to be inactive: their multipliers are null by complementarity)

## Example: the KKT conditions on the QP (2/3)

### Step 3. Complementarity:

- ▶ if the threshold constraint were inactive ( $\lambda = 0$ ) we would have  $x_1 = x_2 = 0$ , which violates  $x_1 + x_2 \geq 6$ : impossible
- ▶ hence the constraint is active:

$$\frac{\lambda}{2} + \frac{\lambda}{4} = 6 \Rightarrow \lambda = 8, \quad \tilde{x}_1 = 4, \quad \tilde{x}_2 = 2, \quad \tilde{f} = 24$$

- ▶ the same numbers found by elementary means, now obtained with the general procedure

## Example: the KKT conditions on the QP (3/3)

### Step 4. Reading of the multiplier:

- ▶ with right-hand side  $6 + \varepsilon$  the optimal value grows by about  $\lambda\varepsilon = 8\varepsilon$
- ▶ verification: with right-hand side  $d$  the solution is  $\tilde{x}_1 = 2d/3$ ,  $\tilde{x}_2 = d/3$ , hence

$$\tilde{f}(d) = \frac{2}{3}d^2 \Rightarrow \tilde{f}(6 + \varepsilon) = 24 + 8\varepsilon + \frac{2}{3}\varepsilon^2 \quad \checkmark$$

# Multipliers and shadow prices

- ▶ The KKT multipliers generalize the shadow prices of the LP: they measure the marginal variation of the optimal value with respect to the right-hand side of the constraint
- ▶ When the duals are not available (general nonlinear models) we estimate the multiplier *by perturbation*:
  - ▶ the right-hand side is increased by  $\varepsilon$
  - ▶ the problem is re-optimized
  - ▶ the incremental ratio is computed
- ▶ This numerical verification is itself a recurring exercise

# Summary: Nonlinear optimization

## Convexity:

- ▶ In a convex problem every local minimum is global
- ▶ A QP is convex if and only if  $\mathbf{Q} \succeq 0$

## KKT conditions:

- ▶ Stationarity of the Lagrangian and complementarity
- ▶ Sufficient in convex problems
- ▶ In the linear case they are complementary slackness
- ▶ The multipliers are shadow prices; they can also be estimated by perturbation

# The sensitivity analysis protocol

# Introduction: The sensitivity protocol

- ▶ Sensitivity analysis is **part of the model**, not an appendix
- ▶ In every laboratory session we follow the same **six-step protocol**
- ▶ We close with three exercises on duality, KKT and convexity

## The six-step protocol (1/2)

- 
- |                                    |                                                                                                                                                                                                                  |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Base scenario                   | solve, verify feasibility, identify the active constraints                                                                                                                                                       |
| 2. One-at-a-time                   | vary one key parameter over a grid; plot value/decisions                                                                                                                                                         |
| 3. Shadow prices and reduced costs | compare the marginal prediction of the dual with a re-optimization after a small perturbation; read the reduced cost $\bar{c}_j$ of every variable at zero: the threshold beyond which it is worth activating it |
-

## The six-step protocol (2/2)

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|              |                                                                                              |
|--------------|----------------------------------------------------------------------------------------------|
| 4. Scenarios | build at least three consistent scenarios (pessimistic, central, optimistic)                 |
| 5. Trade-off | vary a weight or a threshold to trace a frontier (cost-service, risk-return, cost-emissions) |
| 6. Stability | flag abrupt changes in the optimal mix and fragile estimated parameters                      |

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## Exercise: duality

- ▶ In the  $2 \times 2$  LP of the example it is possible to increase  $b_1$  by 10 units by paying 12 per unit. Is it worth it?
- ▶ Up to which unit price is it worth it?
- ▶ Up to which value of  $b_1$  does the shadow price of 14 hold?

*Hint:* recompute the optimal vertex when the first constraint moves to  $90 + \Delta$  and find the  $\Delta$  beyond which the solution changes structure.

# Solution

- ▶ With the first constraint at  $90 + \Delta$  the optimal vertex solves  $x_1 + 3x_2 = 90 + \Delta$ ,  $2x_1 + x_2 = 80$ :

$$\tilde{x}_1 = \frac{150 - \Delta}{5}, \quad \tilde{x}_2 = \frac{100 + 2\Delta}{5}$$

- ▶  $\tilde{x}_1 \geq 0$  as long as  $\Delta \leq 150$ , that is up to  $b_1 = 240$ : up to there the shadow price 14 is exact
- ▶ 10 units are worth  $10 \cdot 14 = 140$  and cost  $10 \cdot 12 = 120$ : it is worth it (net gain 20)
- ▶ It is worth it as long as the unit price is below 14

## Exercise: KKT

Solve with the KKT conditions:

$$\min (x - 3)^2 + (y - 1)^2 \quad \text{subject to} \quad x + 2y \leq 2, \quad x, y \geq 0$$

- ▶ Say which constraints are active at the optimum
- ▶ Interpret the multipliers

## Solution (1/2)

- ▶ The point  $(3, 1)$  is not feasible; its projection onto the line  $x + 2y = 2$  is  $(2.4, -0.2)$ , with  $y < 0$ : hence  $y \geq 0$  will be active too
- ▶ Candidate:  $x + 2y = 2, y = 0 \Rightarrow (\tilde{x}, \tilde{y}) = (2, 0), \tilde{f} = 1 + 1 = 2$
- ▶ Lagrangian:  $L = (x - 3)^2 + (y - 1)^2 + \lambda(x + 2y - 2) - \mu y$

$$\frac{\partial L}{\partial x} = 2(x - 3) + \lambda = -2 + \lambda = 0 \quad \Rightarrow \quad \lambda = 2$$

$$\frac{\partial L}{\partial y} = 2(y - 1) + 2\lambda - \mu = -2 + 4 - \mu = 0 \quad \Rightarrow \quad \mu = 2$$

## Solution (2/2)

- ▶  $\lambda = 2 \geq 0$ ,  $\mu = 2 \geq 0$ ; the constraint  $x \geq 0$  is not active ( $\tilde{x} = 2$ ) and has null multiplier: the KKT conditions hold
- ▶ The problem is convex:  $(2, 0)$  is the global optimum
- ▶ Active:  $x + 2y \leq 2$  and  $y \geq 0$
- ▶  $\lambda = 2$ : relaxing the constraint to  $x + 2y \leq 2 + \varepsilon$  the optimal value decreases by about  $2\varepsilon$
- ▶  $\mu = 2$ : if  $y \geq -\varepsilon$  were allowed the optimal value would decrease by about  $2\varepsilon$

## Exercise: convexity

For which values of the parameter  $\gamma$  is the function

$$f(x_1, x_2) = x_1^2 + \gamma x_1 x_2 + x_2^2$$

convex?

(Write  $\mathbf{Q}$  and impose  $\mathbf{Q} \succeq 0$ .)

## Solution

- ▶ With  $f = \frac{1}{2} \sum_i \sum_j q_{ij} x_i x_j$ :

$$\mathbf{Q} = \begin{pmatrix} 2 & \gamma \\ \gamma & 2 \end{pmatrix}$$

- ▶ A symmetric  $2 \times 2$  matrix is positive semidefinite if and only if the diagonal entries and the determinant are non-negative:

$$2 \geq 0, \quad \det \mathbf{Q} = 4 - \gamma^2 \geq 0$$

- ▶ Hence  $f$  is convex if and only if  $-2 \leq \gamma \leq 2$

# Summary: The sensitivity protocol

- ▶ Sensitivity is part of the model
- ▶ Six steps: base scenario, one-at-a-time, shadow prices and reduced costs, scenarios, trade-off, stability
- ▶ Every marginal prediction must be compared with a re-optimization

# Final summary (1/2)

## Linear programming:

- ▶ Primal and dual, linked by weak and strong duality
- ▶ Complementary slackness as optimality conditions
- ▶ Shadow prices, reduced costs and validity ranges

## Nonlinear optimization:

- ▶ Convexity: every local minimum is global
- ▶ KKT conditions: complementary slackness, generalized

## Final summary (2/2)

### In the laboratory:

- ▶ Shadow prices (and multipliers) are verified by perturbation
- ▶ Every shadow price is quoted with its validity range
- ▶ Every laboratory session follows the six-step sensitivity protocol